Survival Analysis in Genomics: Models, Methods, Key Equations, and Future Directions

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Abstract

Genomics has transformed biomedical research by enabling high-dimensional molecular measurements (e.g., gene expression, somatic mutations, copy-number variation, methylation, proteomics) at cohort scale. Many impactful questions are time-to-event in nature, including overall survival, progression-free survival, time to recurrence, and time to treatment resistance. This review surveys survival analysis methods used in genomics, emphasizing (i) classical semiparametric modeling, (ii) high-dimensional regularization and dimension reduction, (iii) modern machine learning and deep learning, and (iv) practical evaluation issues. We highlight core mathematical formulations that recur throughout the literature and conclude with forward-looking research directions.

1 Introduction

Time-to-event outcomes arise naturally in cancer and complex disease studies where individuals are followed longitudinally and events occur at heterogeneous times, often with right censoring. Genomics adds two defining characteristics: (1) *high dimensionality* where the number of molecular features p can far exceed sample size n, and (2) *biological structure* (pathways, networks, correlated expression modules, multi-omics layers) that can be exploited for prediction and interpretation. The Cox proportional hazards (PH) model is foundational due to interpretability and semiparametric flexibility [5]. However, genomic settings often require regularization, dimension reduction, and non-linear modeling.

Large public resources such as The Cancer Genome Atlas (TCGA) catalyzed method development by providing multi-omics measurements linked to clinical outcomes [26, 21, 14]. Early microarray studies demonstrated that gene expression signatures can stratify patients by recurrence and metastasis risk [23, 16].

2 Notation and data structure (genomics survival)

For individual $i = 1, \ldots, n$, observe:

$$T_i = \text{event time}, \quad C_i = \text{censoring time}, \quad \tilde{T}_i = \min(T_i, C_i), \quad \Delta_i = \mathbf{1}(T_i \le C_i),$$

and high-dimensional genomic covariates $x_i \in \mathbb{R}^p$ (e.g., expression levels, mutation indicators, pathway scores). Define the counting-process style at-risk indicator and event indicator:

$$Y_i(t) = \mathbf{1}(\tilde{T}_i \ge t), \qquad N_i(t) = \mathbf{1}(\tilde{T}_i \le t,\ \Delta_i = 1).$$

3 Classical survival modeling foundations

3.1 Cox proportional hazards model

The Cox PH model posits:

$$h(t \mid x) = h_0(t) \exp(x^\top \beta), \tag{1}$$

where $h_0(t)$ is the baseline hazard and $\beta \in \mathbb{R}^p$ are log hazard ratios.

3.1.1 Partial likelihood

Let $t_{(1)} < \cdots < t_{(D)}$ be distinct event times and $\mathcal{R}(t)$ the risk set just before time t. The Cox *partial likelihood* is

$$L(\beta) = \prod_{d=1}^{D} \frac{\exp(x_{i_d}^\top \beta)}{\sum_{j \in \mathcal{R}(t_{(d)})} \exp(x_j^\top \beta)}, \qquad \ell(\beta) = \log L(\beta), \tag{2}$$

where i_d denotes the individual experiencing the event at $t_{(d)}$. The score (gradient) has the familiar "observed minus expected" structure:

$$U(\beta) = \frac{\partial \ell(\beta)}{\partial \beta} = \sum_{d=1}^{D} \left[x_{i_d} - \frac{\sum_{j \in \mathcal{R}(t_{(d)})} x_j \exp(x_j^\top \beta)}{\sum_{j \in \mathcal{R}(t_{(d)})} \exp(x_j^\top \beta)} \right]. \tag{3}$$

3.1.2 Baseline hazard estimation (Breslow)

After estimating $\hat{\beta}$, the cumulative baseline hazard is commonly estimated by the Breslow estimator:

$$\hat{H}_0(t) = \sum_{t_{(d)} \le t} \frac{1}{\sum_{j \in \mathcal{R}(t_{(d)})} \exp(x_j^\top \hat{\beta})}, \tag{4}$$

yielding individual survival curves:

$$\hat{S}(t \mid x) = \exp\Big(-\hat{H}_0(t) \exp(x^\top \hat{\beta}) \Big).$$

3.2 Why genomics stresses classical modeling

In genomics, p is often 10^3–10^5 while n may be 10^2–10^3, so unconstrained estimation of β is ill-posed. This motivates feature screening, shrinkage/regularization, and dimension reduction.

4 High-dimensional regularization and feature selection

4.1 Penalized Cox regression

The LASSO Cox model solves [20]:

$$\hat{\beta}(\lambda) = \arg\max_{\beta} \left\{ \ell(\beta) - \lambda \|\beta\|_1 \right\}, \tag{5}$$

while elastic net uses a mixture of ℓ_1 and ℓ_2 penalties (often beneficial for correlated genes):

$$\hat{\beta}(\lambda, \alpha) = \arg\max_{\beta} \left\{ \ell(\beta) - \lambda \left(\alpha \|\beta\|_1 + \tfrac{1-\alpha}{2} \|\beta\|_2^2 \right) \right\}, \qquad \alpha \in [0, 1]. \tag{6}$$

Coordinate descent provides scalable regularization paths for Cox models [18].

4.2 Supervised principal components

Supervised principal components (SPC) first selects features associated with the outcome (e.g., via univariate Cox scores), then runs PCA on the selected matrix to obtain low-dimensional components $z_i \in \mathbb{R}^k$ and fits Cox on z_i [2]. A schematic is:

$$x_i \in \mathbb{R}^p \xrightarrow[\text{screening}]{} x_i^{(\mathcal{S})} \in \mathbb{R}^{|\mathcal{S}|} \xrightarrow[\text{PCA}]{} z_i \in \mathbb{R}^k \xrightarrow[\text{Cox}]{} \hat{h}(t \mid z_i).$$

5 Machine learning methods for genomic survival

5.1 Random survival forests

Random survival forests (RSF) build an ensemble of survival trees and average estimated cumulative hazards [10]. Conceptually, for a tree b producing a terminal node $\nu_b(x)$ for covariates x, RSF estimates a node-specific cumulative hazard $\hat{H}_b(t \mid x)$ and averages:

$$\hat{H}_{\text{RSF}}(t \mid x) = \frac{1}{B} \sum_{b=1}^{B} \hat{H}_b(t \mid x), \qquad \hat{S}_{\text{RSF}}(t \mid x) = \exp\Big(-\hat{H}_{\text{RSF}}(t \mid x)\Big). \tag{7}$$

RSF can capture non-linearities and interactions (useful for multi-gene effects), but interpretability typically requires post-hoc tools. In the original RSF development paper, the method was evaluated in both simulation and clinical survival data, with explicit survival-specific splitting rules and out-of-bag (OOB) error estimation tailored to right-censored outcomes [10]. The authors also demonstrated the framework on a coronary artery bypass grafting cohort and emphasized that OOB error closely tracked held-out test error across settings, which is practically useful for high-dimensional genomics workflows where external validation cohorts are limited [10].

5.2 Support vector survival methods

Support vector approaches often frame survival prediction as a ranking problem, aiming to order subjects by risk while handling censoring [22]. These can be competitive in high-dimensional settings, though biological interpretability is typically less direct than sparse Cox models.

6 Deep learning and multi-omics integration

A common deep survival approach is to model a non-linear risk score $f_\theta(x)$ using a neural network, and replace $x^\top \beta$ in the Cox model with $f_\theta(x)$. The *negative partial log-likelihood* loss is:

$$\mathcal{L}(\theta) = -\sum_{i:\Delta_i=1} \left[f_\theta(x_i) - \log\left(\sum_{j \in \mathcal{R}(\tilde{T}_i)} \exp(f_\theta(x_j)) \right) \right], \tag{8}$$

as popularized by DeepSurv [11]. Genomic applications often incorporate multi-omics by concatenation, modality-specific subnetworks, or latent fusion representations, but must manage overfitting when n is modest. Quantitatively, the DeepSurv benchmark results suggest "comparable or better" discrimination in several datasets rather than universal dominance: in METABRIC, reported C-index values were 0.654 (DeepSurv) vs 0.632 (Cox) vs 0.620 (RSF); in SUPPORT, DeepSurv (0.619) and RSF (0.619) were essentially tied and both above Cox (0.583); and in a transfer setup (trained on Rotterdam, tested on GBSG), DeepSurv achieved 0.676 vs 0.659 (Cox) and 0.648 (RSF) [11]. These results support using deep models when nonlinearity is plausible, but also reinforce the need for strong external validation and calibration checks in moderate-n omics cohorts.

7 Competing risks analysis in genomics

In genomic studies, the event of interest is often *not* the only way follow-up can terminate. For example:

- **Cancer-specific death** vs **death from other causes** (especially in older cohorts and long follow-up),
- **Recurrence/progression** vs **death without recurrence** (disease-free survival settings),
- **Relapse** vs **non-relapse mortality** in transplant/immuno-oncology cohorts.

These settings require competing risks methods because treating competing events as independent censoring generally yields biased estimates of event probabilities.

7.1 Cause-specific hazards and the cumulative incidence function

Let $K \in \{1, \ldots, J\}$ denote failure type. The *cause-specific hazard* for cause k is

$$h_k(t \mid x) = \lim_{\Delta t \downarrow 0} \frac{\Pr(t \le T < t + \Delta t,\ K = k \mid T \ge t,\ x)}{\Delta t}. \tag{9}$$

A Cox-type model is often assumed for each cause:

$$h_k(t \mid x) = h_{0k}(t) \exp(x^\top \beta_k),$$

which is attractive in genomics because it aligns with the familiar Cox machinery and allows β_k to be estimated with partial likelihood by treating other causes as censored (but *only* for estimating cause-specific hazards).

The estimand that many clinical genomic questions care about is the *cumulative incidence function (CIF)*:

$$F_k(t \mid x) = \Pr(T \le t,\ K = k \mid x). \tag{10}$$

In general, $F_k(t)$ depends on all cause-specific hazards through

$$F_k(t \mid x) = \int_0^t S(u \mid x)\, h_k(u \mid x)\, du, \qquad S(u \mid x) = \exp\left(- \int_0^u \sum_{\ell=1}^{J} h_\ell(s \mid x)\, ds \right),$$

so modeling a single cause-specific hazard does not directly parameterize F_k.

7.2 Fine–Gray model for the subdistribution hazard

Fine and Gray proposed a proportional hazards model targeting the CIF directly via the *subdistribution hazard* [6]:

$$\tilde{h}_k(t \mid x) = \lim_{\Delta t \downarrow 0} \frac{\Pr(t \le T < t + \Delta t,\ K = k \mid T \ge t \text{ or } (T < t, K \ne k),\ x)}{\Delta t}. \tag{11}$$

The Fine–Gray proportional subdistribution hazards (PSH) model assumes

$$\tilde{h}_k(t \mid x) = \tilde{h}_{0k}(t) \exp(x^\top \gamma_k), \tag{12}$$

where γ_k is interpretable in terms of relative changes to the CIF for cause k. In genomic prognostic modeling, Fine–Gray is often preferred when the scientific target is the absolute risk of a specific event type by time t (e.g., cancer-specific death by 5 years), rather than the instantaneous cause-specific hazard.

7.3 Testing and group comparisons

When comparing CIFs across groups (e.g., molecular subtypes), Gray's test is a standard nonparametric approach [8]. This is a common first step in genomics papers: stratify by subtype/signature, estimate CIFs for each competing event, then proceed to regression modeling.

7.4 High-dimensional competing risks methods relevant to genomics

Competing risks with genomics features inherits the same large-p small-n constraints as standard survival analysis, but now with multiple event processes and competing estimands (cause-specific vs CIF). Several method families are explicitly developed for high-dimensional competing risks and have been used in microarray/genomic contexts:

Boosting for proportional subdistribution hazards. Binder et al. [3] proposed a boosting approach for fitting Fine–Gray PSH models in high-dimensional settings, explicitly motivated by microarray-type predictors and demonstrating how to combine clinical covariates with many molecular features. A technically important contribution is their adaptation of bootstrap .632+ prediction-error estimation to competing-risks prediction curves, which gives a practical internal-validation strategy when $p \gg n$ and repeated model fitting is required [3].

Penalized regression and computational scaling. Penalized Fine–Gray models (e.g., LASSO/elastic net) have been studied with a focus on scalable fitting and screening rules; for example, PSH-SAFE provides safe feature elimination for penalized Fine–Gray regression [25]. These tools are practical when p is large and competing risks regression must be re-fit repeatedly during cross-validation.

Inference and stability in high dimensions. Beyond point estimation, uncertainty quantification is an active area. Hou et al. [9] develops one-step bias-corrected inference for high-dimensional Fine–Gray regression, enabling confidence intervals and hypothesis testing for selected effects even in $p > n$ regimes. Their application to non-cancer mortality among prostate cancer patients illustrates why this matters in practice: genomic and clinical signals can have different roles for incidence prediction vs inferential interpretation [9]. For stable feature selection under correlated predictors (typical in genomics), methods such as Random Approximate Elastic Net (RAEN) have been proposed within the PSH framework [19].

7.5 Practical guidance for genomics writeups: choosing the estimand

A helpful rule of thumb is:

- Use **cause-specific Cox** when interpreting biological mechanisms of instantaneous risk for each event type (and when modeling each cause is itself the target).
- Use **Fine–Gray** when predicting or comparing **absolute risk** (CIF) of a particular event type by a clinically meaningful horizon.

Many genomics studies benefit from reporting both: cause-specific hazard ratios (mechanistic interpretation) and CIF-based predictions (clinical risk).

8 Evaluation metrics that recur in genomic survival papers

8.1 Concordance index (C-index)

The C-index measures the fraction of comparable pairs where predicted risk is correctly ordered. Let η_i be a risk score (larger means higher risk). A common estimator is:

$$\widehat{C} = \frac{\sum_{i \neq j} \mathbf{1}(\tilde{T}_i < \tilde{T}_j,\ \Delta_i = 1)\, \mathbf{1}(\eta_i > \eta_j)}{\sum_{i \neq j} \mathbf{1}(\tilde{T}_i < \tilde{T}_j,\ \Delta_i = 1)}. \tag{13}$$

In genomics, the C-index is popular but can be optimistic if tuning/feature selection is not nested within validation.

8.2 Time-dependent AUC with inverse probability of censoring weights (IPCW)

For a fixed horizon t, one can define "cases" as those with $T \leq t$ and "controls" as those with $T > t$, and weight observations to adjust for censoring. A typical IPCW weight uses the censoring survival function $\hat{G}(\cdot)$:

$$w_i(t) = \frac{\mathbf{1}(\tilde{T}_i \leq t,\ \Delta_i = 1)}{\hat{G}(\tilde{T}_i)} \quad \text{or} \quad w_i(t) = \frac{\mathbf{1}(\tilde{T}_i > t)}{\hat{G}(t)},$$

depending on the chosen definition. The key theme is that censoring-adjusted metrics are preferred when censoring is substantial and heterogeneous.

9 Applications and data resources

9.1 From discovery signatures to clinical assays

Early gene expression studies showed that prognostic signatures can stratify breast cancer patients by risk of distant metastasis [23]. Subsequent work translated multigene signatures into clinical assays used to guide treatment decisions; a prominent example is the 21-gene recurrence score for tamoxifen-treated, node-negative breast cancer [16]. These case studies underscore (i) strong internal validation discipline and (ii) the necessity of external cohort replication.

9.2 Pan-cancer modeling with TCGA

TCGA and related resources are widely used for benchmarking and pan-cancer prognostic modeling [26, 21]. Because endpoints and follow-up differ by cancer type, harmonized endpoint definitions (e.g., TCGA-CDR) and careful attention to missingness and covariate shift are critical for high-quality survival analytics [14].

10 Left truncation and right censoring in genomic cohorts

Many genomic survival datasets (especially biobank-scale resources and retrospective clinical cohorts) exhibit *right censoring* and may also exhibit *left truncation* (delayed entry). Left truncation occurs when individuals enter the risk set only after surviving to an entry time E_i (e.g., time from diagnosis to sequencing/consent), so the observed sample is conditional on $T_i \geq E_i$ [12]. Ignoring left truncation typically biases hazard estimates and can distort genomic effect sizes, particularly when entry depends on disease severity or healthcare access.

10.1 Data structure with delayed entry

With delayed entry, each subject has an entry time E_i and the at-risk indicator becomes

$$Y_i(t) = \mathbf{1}(E_i \le t \le \tilde{T}_i), \tag{14}$$

while right censoring remains encoded via $(\tilde{T}_i, \Delta_i)$ as before. In genomics, E_i is often the time of biospecimen availability, sequencing, or enrollment into a molecular profiling program, which can be substantially after diagnosis.

10.2 Cox partial likelihood with left truncation

The Cox partial likelihood extends directly by redefining the risk set to include only individuals who have entered by t and have not yet had an event/censoring:

$$\mathcal{R}(t) = \{j : \ E_j \le t \le \tilde{T}_j\}.$$

Thus the partial likelihood retains the same form as (2), but with the delayed-entry risk set:

$$L(\beta) = \prod_{d=1}^{D} \frac{\exp(x_{i_d}^\top \beta)}{\sum_{j \in \mathcal{R}(t_{(d)})} \exp(x_j^\top \beta)}. \tag{15}$$

This is the standard approach implemented in mainstream software (e.g., `Surv(entry, exit, event)` in `R`).

10.3 Counting-process view and implications for omics pipelines

Under a counting-process formulation, delayed entry modifies $Y_i(t)$ but keeps the martingale structure used for asymptotic theory and robust variance estimation [1]. Practically, the key implications for genomic survival modeling are:

- **Preprocessing and eligibility**: if molecular data are available only for those who survive to profiling, the analytic cohort can be enriched for longer survivors. This is common in retrospective tumor sequencing and can create "immortal time"-like artifacts unless entry times are modeled.
- **Feature selection and validation**: left truncation must be respected inside cross-validation folds (risk sets and baseline hazard estimation depend on E_i). Leakage can be more subtle when folds have different entry-time distributions.
- **Multi-omics and longitudinal sampling**: when omics are collected at multiple times, each measurement can induce its own "entry" (time-varying covariates). Landmarking or joint models may be preferable when omics are only observed after certain survival milestones.

10.4 When to worry most in genomics

Left truncation is most severe when (i) sequencing/enrollment occurs long after diagnosis, (ii) entry time is associated with prognosis or care pathways, or (iii) the study is assembled from survivors (prevalent cases). In these settings, robust reporting should include the distribution of entry times, the fraction of subjects with delayed entry, and sensitivity analyses comparing models with and without truncation adjustment [12].

10.5 High-dimensional methods under left truncation: what changes mathematically

From a modeling standpoint, most high-dimensional survival methods extend to left truncation by *the same principle*: keep the model/loss, but compute it using the delayed-entry risk set $\mathcal{R}(t) = \{j : E_j \le t \le \tilde{T}_j\}$. For example, the LASSO Cox objective becomes

$$\hat{\beta}(\lambda) = \arg\max_{\beta} \left\{ \sum_{i:\Delta_i=1} \left[x_i^\top \beta - \log \left(\sum_{j:\, E_j \le \tilde{T}_i \le \tilde{T}_j} \exp(x_j^\top \beta) \right) \right] - \lambda \|\beta\|_1 \right\}. \tag{16}$$

Algorithmically, coordinate descent and other convex-optimization routines are unchanged, but the bookkeeping of risk sets (and quantities derived from them) must reflect entry times.

A practically important takeaway is that *model selection must also be truncation-aware*: cross-validation should compute partial-likelihood deviance using delayed-entry risk sets in each fold, otherwise tuning can implicitly optimize the wrong target.

10.6 Clinico-genomic evidence: penalized Cox with delayed entry

Left truncation is not only a theoretical nuance—it has been shown to materially affect both estimation and prediction in high-dimensional, real-world clinico-genomic survival modeling. McGough et al. [15] studied penalized Cox models (including LASSO) under left-truncated and right-censored follow-up and demonstrated that failure to adjust for truncation can induce substantial bias and miscalibration; their paper also provides a clear simulation framework and a motivating analysis in a high-dimensional clinico-genomic database. In their stage IV NSCLC clinico-genomic analysis ($n = 4429$), 21.5% of patients had delayed entry greater than one year, and the correlation between diagnosis year and truncation time was strongly negative ($\rho = -0.63$), quantitatively highlighting that delayed entry is structured rather than random [15].

Complementarily, Brown et al. [4] focuses on the *selection bias mechanism* in clinical genomic studies: genomic profiling often occurs months/years after diagnosis, so those who die or are lost before profiling are systematically excluded. In stage IV NSCLC and colorectal cancer examples, they showed that naïve analyses that ignored delayed entry overestimated median overall survival by more than one year relative to risk-set-correct analyses, directly demonstrating the magnitude of this bias in clinicogenomic settings [4].

10.7 Beyond Cox: trees/forests and deep models for LTRC data

While Cox-based methods dominate genomics due to interpretability and software maturity, delayed entry also arises in settings where non-linear models are attractive (e.g., heterogeneous treatment effects, high-order interactions). Tree-based methods have been explicitly generalized to left-truncated right-censored (LTRC) data; for instance, Fu and Simonoff [7] proposes survival tree constructions for LTRC data and connects them to time-varying covariate survival trees. More recently, survival forests for LTRC data have been developed [13]. On the deep-learning side, Seo and Lee [17] proposes a neural-network approach designed to incorporate left truncation alongside right censoring in cancer survival settings.

10.8 What to emphasize in a genomics writeup

If you want the left-truncation section to feel "genomics-native," the most compelling points to highlight are:

- **The origin time vs profiling time mismatch**: diagnosis/treatment start is the scientific time origin, while sequencing is the entry time.
- **Truncation-aware evaluation**: tuning and performance estimation must be done with delayed-entry risk sets inside CV folds.
- **Sensitivity analyses**: report entry-time distributions and rerun key models under different plausible origins (diagnosis vs treatment line start), especially in real-world clinico-genomic databases.

11 Practical issues (pitfalls) in genomic survival modeling

11.1 Leakage and "double dipping"

If screening/normalization/gene selection uses all samples before cross-validation, performance estimates are biased upward. Best practice is *fully nested* resampling where every preprocessing step that uses outcome information is performed inside each training fold [24].

11.2 Batch effects and confounding

Omics measurements can contain technical variation correlated with site, sequencing batch, or processing date. If these factors also correlate with prognosis or treatment, batch effects can mimic biological signals. This makes transparent preprocessing and sensitivity analyses essential.

11.3 Endpoint harmonization and label quality

Cross-cohort genomic survival studies are also sensitive to outcome-definition drift. The TCGA-CDR harmonization effort curated survival endpoints for 11,160 patients across 33 cancer types and reported four standardized outcomes (OS, PFI, DFI, DSS), explicitly to reduce cross-study inconsistencies in endpoint construction [14]. Practically, if one cohort uses loosely defined progression while another uses stricter radiologic progression, apparent model transportability can degrade for reasons unrelated to biology. For this reason, endpoint-definition documentation should be treated as part of model specification, not merely dataset metadata [14].

11.4 Interpretability and stability

Sparse Cox is interpretable (β maps to hazard ratios), but correlated genes can make selected sets unstable. Elastic net and pathway/group penalties can improve stability [18]. For RSF and deep models, interpretation requires post-hoc summaries (importance, partial dependence), ideally at pathway/module level.

12 Future directions

Below are research directions that appear especially promising for "survival analysis in genomics" and fit naturally into an advanced survival analysis course discussion.

Table 1: Evidence snapshots from representative studies used in this review.

Study	Data context	Endpoint context	Model class	Key quantitative takeaway
Ishwaran 2008	Simulations + coronary artery bypass grafting cohort	Right-censored survival	Random survival forests	OOB prediction error closely tracked test error across settings; demonstrated robust survival-forest workflow for censored outcomes [10].
Katzman 2018	METABRIC, SUPPORT, Rotterdam/GBSG	Overall survival/time-to-event benchmarks	Deep Cox neural network	C-index: METABRIC 0.654 (DeepSurv) vs 0.632 (Cox) vs 0.620 (RSF); transfer Rotterdam→GBSG 0.676 vs 0.659 vs 0.648 [11].
Binder 2009	High-dimensional microarray + clinical covariates	Competing risks CIF prediction	Boosted Fine–Gray PSH	Adapted bootstrap .632+ prediction-error curves for competing-risks validation in high-dimensional settings [3].
McGough 2021	Stage IV NSCLC clinicogenomic cohort ($n = 4429$)	Left-truncated, right-censored survival	Penalized Cox (LTRC-aware)	21.5% had delayed entry > 1 year; diagnosis year and truncation time had $\rho = -0.63$, motivating explicit delayed-entry adjustment [15].
Brown 2022	Clinicogenomic NSCLC/CRC cohorts	Overall survival with delayed profiling entry	Bias assessment under delayed entry	Ignoring delayed entry overestimated median OS by > 1 year in stage IV NSCLC/CRC examples [4].
Liu 2018	TCGA-CDR (33 cancers, 11,160 patients)	Harmonized OS/PFI/DFI/DSS endpoints	Data harmonization for survival analytics	Large-scale endpoint harmonization reduced ambiguity in pan-cancer prognostic analyses and enabled more consistent cross-cancer benchmarking [14].

12.1 (1) Transportability and external validity across cohorts

Genomic predictors often degrade when moved across studies due to differences in platforms, processing pipelines, population composition, treatment standards, and censoring patterns. Methodological gaps include:

- domain adaptation for censored outcomes (including recalibration of baseline hazards),
- evaluation protocols that separate *within-cohort* CV from *cross-cohort* generalization,
- explicit modeling of covariate shift and label shift in survival settings.

12.2 (2) Competing risks and multi-state models in molecular oncology

Many cancer datasets involve competing events (e.g., cancer-specific vs other-cause mortality) or multi-state trajectories (diagnosis → progression → death). Genomic studies frequently simplify to OS, but richer modeling could leverage:

- cause-specific hazards and subdistribution hazards,
- transition-specific genomics effects in multi-state models,
- joint modeling of longitudinal biomarkers and survival.

12.3 (3) Dynamic prediction with longitudinal genomics

Increasingly, genomics is measured repeatedly (liquid biopsies, serial tumor biopsies, longitudinal single-cell). This motivates:

- landmarking and time-varying coefficient models,
- joint models linking evolving molecular markers to hazard,
- representation learning for irregularly sampled longitudinal omics.

12.4 (4) Integrating biological structure: pathways, networks, and priors

Genes operate in pathways and interaction networks, suggesting structured penalties and Bayesian priors:

- group LASSO / sparse group penalties aligned to pathways,
- graph-guided fused penalties to encourage smoothness over gene networks,
- hierarchical Bayesian survival models borrowing strength across related genes or modules.

A key open question is how to preserve interpretability while improving stability and reproducibility.

12.5 (5) Uncertainty quantification for high-dimensional and deep survival models

Genomic survival models are often used for decision support, where uncertainty matters. Promising directions include:

- post-selection inference for penalized Cox,
- Bayesian neural survival models with calibrated predictive intervals,
- conformal prediction ideas adapted to censoring (for time-to-event risk sets).

12.6 (6) Robustness to endpoint noise and label quality

Clinical endpoints (especially PFS/DFS) can be inconsistently recorded across institutions. Harmonization efforts like TCGA-CDR help [14], but future work could incorporate:

- measurement-error models for event times,
- sensitivity analyses for endpoint definition variation,
- semi-supervised approaches that exploit unlabeled/weakly labeled survival data.

13 Conclusion

Survival analysis in genomics sits at the intersection of classical time-to-event modeling and modern high-dimensional learning. Core formulations—Cox partial likelihood, penalized objectives, and censoring-aware metrics—underpin much of the literature, while RSF and deep survival models expand expressivity for multi-omics integration. Future progress will likely depend on cross-cohort generalization, richer event-process modeling (competing risks/multi-state), principled uncertainty quantification, and better integration of biological structure.

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